

Samya Aggarwal

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EDUCATION

Thapar Institute of Engineering and Technology
BE in Computer Science and Business Systems
CGPA: 9.74/10.0

Patiala, Punjab
Sep 2023 – May 2027

Cambridge International School (CBSE)
Class XII (CBSE): 95.16%

Amritsar, Punjab
May 2022 – May 2023

SKILLS

Languages: Python, C++, C, JavaScript, SQL
Web: Django, Django REST Framework, Node.js, FastAPI, Django Channels, WebSockets
Python Libraries: Pandas, NumPy, Matplotlib, SHAP
Tools/Cloud: Git/GitHub, Redis, AWS, Linux
Databases: MySQL, PostgreSQL, SQLite
Concepts: Concurrency, Multithreading, Distributed Systems, Agile Development, CI/CD, REST APIs
Coursework: Data Structures and Algorithms (DSA), Database Management Systems (DBMS), Object Oriented Programming (OOP), Operating Systems (OS), Computer Networks

PROFESSIONAL EXPERIENCE

Zylectra **Venture Lab Thapra**
Software Development Intern *Sep 2025 – Oct 2025*

- Built scalable Node.js backend services handling concurrent booking requests with real-time conflict-resolution logic.
- Resolved API failures and performance bottlenecks using logging, profiling and structured debugging approaches.
- Integrated ML-based recommendations via FastAPI and improved API response latency by ~20% by introducing Redis caching.

InveshoAI **Remote (US-Based)**
Software Engineering Intern *Jun 2025 – Aug 2025*

- Built a FastAPI-based automation service to streamline internal workflows, structuring reusable functions for scalable task execution.
- Engineered a lightweight comment-generation pipeline using LLM APIs to support automated engagement workflows.
- Built and deployed internal tooling supporting a Product Hunt launch, contributing to user acquisition across 30+ countries.

PROJECTS

SyncRoomies: Roommate Selection Platform [GitHub](#)

- Designed and deployed a roommate matching platform serving 30+ users with real-time chat and profile matching.
- Architected real-time messaging infrastructure using Django Channels & WebSockets, supporting concurrent interactions across a distributed, fault-tolerant system.
- Integrated Razorpay payment gateway with secure payment verification and transaction management; validated payment flows across 20+ functional and edge-case scenarios.

SafeSpace: AI-Powered Mental Health Platform [GitHub](#)

- Designed a scalable and maintainable FastAPI backend for real-time multimodal inference with 85% model accuracy.
- Engineered a custom ESP32-based wearable with 4 biosensors (ECG, EDA, EMG, Temp) for continuous data streaming.
- Deployed the solution on Vercel with a FastAPI backend, adding a SHAP-based XAI layer for transparency.

SmartKheti: AI-Powered Crop Disease Detection [GitHub](#)

- Designed and deployed a scalable Django backend for image upload, disease prediction, and crop recommendations.
- Integrated CNN and Autoencoder inference pipelines into backend services, achieving 80% disease classification accuracy.
- Developed a backend recommendation module supporting 20+ crops using climate and regional data.

ACHIEVEMENTS & LEADERSHIP

- Amazon ML Summer School 2025:** Selected among **3,000 students** for an intensive machine learning program.
- Reliance Foundation Scholar:** Selected as one of 5,000 students nationwide; earned a **Rs. 2 lakh** merit scholarship.
- Hackathons:** Awarded **3rd Prize** at the **Indian-Israeli Hackathon**; selected for **HackPrinceton'26**.
- Academic Excellence:** Secured **3rd rank** in first and second year (2023–25); awarded a **50% tuition fee** scholarship.
- LeetCode:** Solved 300+ DSA problems across Graphs, DP, Trees, Greedy and Advanced Data Structures.